

Review Article

Efficacy of resistance training with elastic bands compared to proprioceptive training on balance and self-report measures in patients with chronic ankle instability: A systematic review and meta-analysis

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ABSTRACT

Background: Proprioceptive training and resistance training are physiotherapy treatment methods for Chronic Ankle Instability (CAI).

Objective: To compare the efficacy of proprioceptive training to resistance training with elastic bands for treating CAI as measured by the Star Excursion Balance Test (SEBT), the Foot and Ankle Ability Measure (FAAM), and the Cumberland Ankle Instability Tool (CAIT).

Method: Our systematic study and meta-analysis was based on the PICOS and PRISMA protocols. The PubMed, PEDro, and ScienceDirect databases were searched for randomized clinical trials on proprioceptive and resistance training. Risk of bias was assessed according to Cochrane guidelines and quality of evidence was reported using the Grading of Recommendations Assessment, Development, and Evaluation approach (GRADE).

Results: Five studies involving 259 patients were included in the review. According to the findings of the meta-analysis, proprioceptive training was similarly effective with resistance training in SEBT and FAAM measures. Compared with resistance exercise, proprioceptive training demonstrated some benefits in CAIT scores (weighted mean difference [WMD] = −2.21, 95% CI = −4.05–0.36), but these intervention results were not clinically significant (MDC, MCID score >3 points).

Conclusion: Low-quality evidence from studies showed that neither of the interventions was superior on the SEBT or the FAAM scores in individuals with CAI because no clinically significant differences were found. More high-quality studies comparing the two interventions are needed to draw firm conclusions.

1. Introduction

A sprain in the lateral collateral ligament of the ankle joint is a common injury in both professional and amateur athletes and accounts for 80% of ankle injuries (Delahunt et al., 2010). Approximately 40% of people who have experienced an ankle sprain are prone to re-injury, which may lead to Chronic Ankle Instability (CAI) (Hertel & Corbett, 2019; Osborne & Rizzo, 2003). Recurrent ankle sprains cause damage to the joints, muscles, and mechanical receptors and often lead to chronic instability (Doherty et al., 2017; Kim et al., 2019; Osborne & Rizzo, 2003). According to the International Ankle Consortium (Gribble et al.,

2014), CAI is the most used term to describe participants who report certain and ongoing insufficiencies of the ankle joint complex after an acute ankle injury. Components of CAI include functional instability, which is defined as the feeling of ankle joint giving way and feelings of instability and recovery from a “rolling-over” incident. Symptoms of CAI include pain, instability, and loss of function, while athletes often feel that their joint is giving up on them (Delahunt et al., 2010; Hiller et al., 2011; Kosik et al., 2017; Martin et al., 2013; Tsikopoulos et al., 2018; Webster & Gribble, 2010). These symptoms may last for more than 12 months and in some cases for more than two years (Attenborough et al., 2014; Hertel & Corbett, 2019).

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The most common outcome measures used for assessing balance, performance, and symptoms in individuals with CAI are the Star Excursion Balance Test (SEBT), the Foot and Ankle Ability Measure (FAAM), and the Cumberland Ankle Instability Tool (CAIT). The SEBT has the lowest cost and is the most straightforward measure used to assess those with CAI. The tool measures reach range in the anterior (Ant), posteromedial (PM), and posterolateral (PL) directions, with a minimal detectable change (MDC) of 4.9 cm for A, 5.2 cm for PM, and 5.4 cm for PL (Schaefer & Sandrey, 2012).

The Foot and Ankle Ability Measure (FAAM) measures general self-reported functional ability in people with foot, ankle, and leg musculoskeletal injuries (Martin et al., 2005). It is a questionnaire that includes an eight-item sub-scale for athletic performance and a 21-item sub-scale for activities of daily living (ADL), both of which are scored on a 0%–100% scale. A lower percentage indicates a lower level of functional ability (Martin et al., 2005). For the ADL and Sports subscales, the minimal detectable change was 5.7 and 12.3 points respectively, based on a 95% confidence interval (Martin et al., 2005). In contrast, the minimal clinically important differences for the ADL and Sports subscales were 8 and 9 points respectively (Martin et al., 2005).

CAIT is another valid and reliable instrument that can identify the severity of CAI and functional deficits (Hiller, Refshauge, Bundy, Herbert, & Kilbreath, 2006a). It consists of nine questions scored on a 30-point scale with lower scores indicating decreased stability with an MDC score of 3.08 and an MCID score of more than three points (Wright et al., 2017a,b).

Exercise is the cornerstone of CAI rehabilitation (Hall, Docherty, Simon, Kingma, & Klossner, 2015a; Kim & Jeon, 2016). Research has shown that therapeutic exercise in athletic populations with CAI increases strength (Hall et al., 2018a; Smith et al., 2012), dynamic balance (Hall et al., 2018a; E. Lee et al., 2019; Wright et al., 2017a,b), functional ability (Hall, Docherty, Simon, Kingma, & Klossner, 2015a; Hall et al., 2018a; Wright & Linens, 2017), quality of life (Hall et al., 2018b), Hall et al., 2018bnd reduces risk of injury (Attenborough et al., 2014; Loudon et al., 2008; McGrath et al., 2017). Two of the most common types of CAI rehabilitation training approaches are resistance and proprioceptive exercise (Cain et al., 2020; Cruz-Diaz et al., 2015; Hall et al., 2018a; Kim, Estepa-Gallego, Estudillo-Martínez, Castellote-Caballero, & Cruz-Díaz, 2022a; Lapanantasin et al., 2022; Wright et al., 2017a,b).

Proprioception is the neurological process by which the body receives sensory input from its surroundings and integrates it to produce a muscular response (Rivera et al., 2017).

Proprioceptive training refers to any type of training that aims to improve balance (static and dynamic) by developing proprioceptive receptors that positively affect the stability of a joint during static and dynamic functional tasks (Hale et al., 2007). Proprioceptive protocols include dynamic progressive activities such as hop-to-stabilization exercises (Hall et al., 2018a), single-limb stance activities on an unstable surface (Cain et al., 2020; Cruz-Diaz et al., 2015; Hall et al., 2018a; Kim et al., 2022a), weight-bearing positions on an unstable surface (Cain et al., 2020; Cruz-Diaz et al., 2015), walking meditation (Lapanantasin et al., 2022), and wobble board training (Cain et al., 2020; Wright et al., 2017a,b).

Resistance training is applying effort to overcome resistance, which results in increased muscle fiber recruitment and stronger synchronization, ultimately enhancing neuromuscular control and leading to muscular growth (Ahern et al., 2021a). Elastic resistance training is not only the cheaper intervention but is also able to promote similar strength gains to conventional resistance training (Lopes et al., 2019; Ribeiro et al., 2009). Elastic bands are often offered to physiotherapists as not all physiotherapy centers can afford expensive equipment.

The purpose of this review was to investigate the efficacy of two different types of training, resistance band strength and proprioceptive, on improving dynamic postural control and functionality, as measured by the SEBT, FAAM, and CAIT in populations with CAI. The need for a systematic review on this topic is great as these two interventions are the

most used by physiotherapists. Furthermore, conducting such a systematic review could provide clarity on which of the two interventions is most appropriate for a rehabilitation program performed at home. The recent coronavirus COVID-19 pandemic and the relevant restrictions (difficulty to visit physiotherapy centers) solidified the importance of at-home rehabilitation programs without equipment. Finally, we performed a literature search that did not yield any similar studies on the topic, which reinforced our decision to proceed with our research.

2. Method

2.1. Design

Our systematic review was performed in accordance with PRISMA 2020 guidelines (Page et al., 2021). The international Prospective Register of Systematic Reviews (PROSPERO) registration number was CRD42023428769.

2.2. Search strategy

Separate searches were conducted during the period November 2022–March 2023. A systematic, non-time-limited literature search was performed in the online databases PubMed, PEDro, and ScienceDirect, which was completed on March 18, 2023. The search strategy used is depicted in Table 1.

2.3. Eligibility criteria

The selection criteria for our systematic review were developed according to the PICOS framework (Moher et al., 2009). When creating a search strategy for the archive databases and selecting articles, the following factors were taken into consideration:

Population: Individuals with acute or chronic ankle ligament injuries of any grade were eligible. Patients with a history of at least 1 significant ankle sprain in the past 12 months, patients with a history of the previously injured ankle joint “giving way” and/or recurrent sprain and/or “feelings of instability” and Cumberland Ankle Instability Tool (CAIT): <24 (Gribble et al., 2014). Participants were male and female as well as athletic and non-athletic to enable the generalization of results to diverse groups.

Intervention: The intervention consisted of any form of resistance training with elastic bands or elastic tubes targeting the ankle joint. To this end, the search strategy included the following keywords: resistance exercise OR resistance training AND elastic bands OR elastic tubes.

Control: The only comparison groups that could be used were those that received a proprioceptive exercise program. Therefore, studies that had at least one control group consisting of proprioceptive exercise program were included. The search strategy included the following keywords: proprioceptive training OR proprioceptive exercise OR proprioceptive program.

Outcomes: The main and primary outcome measures required in the included studies were balance and functional ability. Included trials were required to report the SEBT, FAAM, and CAIT scores as a primary or secondary outcome.

Table 1
Search strategy.

Database	Search strategy
PubMed ScienceDirect PEDro	(chronic ankle instability OR CAI) AND (resistance band exercise OR resistance band tubes training) AND (proprioceptive exercise OR proprioceptive training) AND (star excursion balance test OR SEBT) AND (Foot and Ankle Ability Measure OR FAAM)

2.4. Studies selection

Randomized clinical trials were included in our review with at least one group receiving resistance training and another following a proprioceptive exercise program. Treatments were given to patients with CAI, while patients not having been diagnosed with CAI were excluded from the review. The following were the inclusion criteria: (1) the entire population regardless of age with CAI, (2) a group receiving any form of resistance exercise program with elastic bands or elastic tubes, (3) a comparison group following a proprioceptive program, (4) dynamic postural control measured with SEBT, (5) functional ability measured with FAAM, and (5) a CAIT score <27. The exclusion criteria were as follows: (1) studies with a semi-experimental design (i.e., they were not randomized clinical trials), (2) articles not available in full-text, (3) articles not published in English, (4) patients with other conditions that affect balance (ear infections, neurological impairments etc.), and (5) previous fixation surgery. The eligibility criteria for the included studies are summarized in Table 2.

2.5. Screening, selection process, and data extraction

Two reviewers independently examined the studies that resulted from the search process. First, using the Mendeley tool, duplicate studies were removed and the titles and abstracts of the remaining studies were evaluated for appropriateness. If a study met the criteria, its full text was retrieved and assessed regarding its suitability. The reviewers had to agree on a set of criteria to include an article in the review. When they could not reach a consensus, a third researcher was asked to aid in their decision. Data from each trial were extracted by two authors into a Excel spreadsheet that included study design, sample size, interventions used, outcomes, and follow-up periods. Finally, each item on the data extraction form needed to be agreed upon by the two reviewers. In case of a dispute, the final call was again made by a third reviewer.

2.6. Quality assessment

Two members of the research team used the Cochrane Risk of Bias (RoB) assessment tool (De Morton, 2009) and the PEDro scale (Higgins et al., 2011) to evaluate the methodological quality and risk of bias in the included studies. The RoB tool includes the following types of bias: selection bias (randomization sequence generation, allocation concealment), performance bias (blinding participants, blinding therapists), detection bias (blinding outcome assessor), attrition bias (incomplete outcome data), reporting bias (source of funding bias or selective outcome reporting), and other bias (sample size). The Cochrane Collaboration tool evaluated each item as low risk, high risk, or unclear (Higgins et al., 2011). All crucial areas must have a low risk of bias for an article to be labeled as low risk. The article was designated as having an unclear risk of bias if all crucial parts were found to have a low or unknown risk of bias. The included study was found to have a high risk of

bias if one or more of its crucial areas were rated as high risk. The quality of the included studies was evaluated with the PEDro scale (Verhagen et al., 1998), which is considered a valid measurement scale for the methodological assessment of clinical trials included in a systematic review (De Morton, 2009; Maher et al., 2003). The PEDro scale is based on 11 criteria, 10 of which contribute to the overall score of a study. The first criterion was not included in the score as it relates to the external validity of the study. The scale assesses the following criteria: random allocation, concealed allocation, between-group similarity at baseline, participant blinding, therapist blinding, assessor blinding, dropout rate, intention-to-treat statistical analysis, between-group statistical comparison, and point measures, and variability data. According to Cashin and McAuley (2020), if the authors of a study report an overall PEDro scale score of 0–3 this is considered “poor”, 4–5 “moderate”, 6–8 “good”, and 9–10 “excellent”. However, it is important to note that these classifications have not been validated. Furthermore, for trials evaluating complex interventions (e.g., exercise) an overall PEDro score of eight out of 10 is optimal (Cashin & McAuley, 2020).

2.7. Quality of evidence assessment

The criteria for evaluating the quality of evidence presented by the studies in our review are in accordance with the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system that the Cochrane Back and Neck (CBN) group included in their 2015 systematic review guidelines (Granholm et al., 2019). The assessment was performed independently by two authors, with a third author available if the two authors could not reach agreement. The quality of evidence was classified as high, moderate, low, or very low based on the presence of study limitations (downgraded if the bias according to RoB was large), indirectness of evidence, inconsistency of results, unexplained heterogeneity (downgraded if significant heterogeneity was presented by visual inspection of $I^2 > 40\%$ and $I^2 > 60\%$), imprecision of results (downgraded if there were fewer than 142 participants; rule of thumb was used) and high probability of publication bias (Granholm et al., 2019). The quality of evidence was classified as high when all elements were negative, moderate when one element involved a serious risk, low when two to three elements showed a serious risk or one to two elements showed a very serious risk, or very low when all elements showed a serious risk or more than two elements showed a very serious risk.

2.8. Data synthesis and analysis

Our meta-analysis was performed using the statistical software Review Manager (RevMan version 5.4). Data synthesis was categorized by group according to the SEBT as anterior-reach direction, posteromedial-reach direction, and posterolateral-reach direction if data were available from the FAAM-ADL, FAAM-Sport, and CAIT. The data extracted from each study were the sample size, mean value, and standard deviation of each variable. It is important to note that when the study contained confidence intervals instead of standard deviations, they were converted to standard deviations according to the Cochrane Handbook. The mean score difference between the pre- and post-intervention periods served as the basis for calculating the study’s outcome. The weighted mean differences (WMDs) and SD with 95% confidence interval (CI) were calculated for continuous data. We used weighted mean difference as it has been reported that this allows the evaluation of studies that use same outcome measures to evaluate the desired outcome (Andrade, 2020). Heterogeneity of studies was assessed using the I^2 statistics model. Due to high heterogeneity a random effects model was adopted. The Cochrane Group has established the following interpretation of the I^2 model: 0–40% heterogeneity may not be relevant or significant, 30–60% indicates moderate heterogeneity, 50–90% indicates substantial heterogeneity, and 75–100% indicates significant heterogeneity (Deeks et al., 2019).

Table 2
Eligibility criteria.

Inclusion criteria	Exclusion criteria
• Study design: randomized control trials • Population of interest: Any gender, any age with CAI • Intervention: Proprioceptive training (e.g., hop-to-stabilization exercises, single-limb stance activities on an unstable surface, weight-bearing positions on an unstable surface, and wobble board training) • Control group: Strength training (e.g., resistance bands, resistance tubes)	• Studies with a semiexperimental design (i.e., they were not randomized clinical trials) • Articles not available in full-text • Articles not published in English • Patients with other conditions that affect balance (ear infections, neurological impairments etc.) • Previous fixation surgery

3. Results

A total of 2952 articles related to our topic resulted from the search process. After removing duplicates, 1810 studies remained. Furthermore, after a thorough review of their titles and abstracts, eight related articles remained for further full-text screening. The full article was then accessed and three articles were removed as one of them had no clear comparison between proprioceptive and resistance training (Donovan et al., 2016a), another did not use proprioceptive training (Hall et al., 2015a), and another used resistance band as balance exercise and not as strength exercise (Lapanantasin et al., 2022). Finally, five articles were included in the qualitative and quantitative analysis of our review (Cain et al., 2020; Cruz-Diaz et al., 2015; Hale et al., 2007; Hall et al., 2018a; Wright et al., 2017a,b). The study selection flow diagram is shown in Fig. 1.

3.1. Characteristics of studies

Participant characteristics of the included studies are listed in Table 3. In total, 259 participants were included: 128 males and 131 females, with a mean age of 22 years. All the studies' populations were described as having chronic or recurring ankle instability. Every study reported participants with unilateral chronic ankle instability and none with bilateral ankle instability. Regarding participation in sports, it was not mentioned what frequency of exercise participation was in Hall et al. (2018a), unlike Cruz-Diaz et al. (2015) and Kim et al. (2022a), whose participants were athletes. Cain et al. (2020) defined athletic activity as taking part in at least 5 h of moderate activity each week and Wright et al. (2017a,b) as participating in more than 1.5 h of moderate-vigorous physical activity per week. The methods used to follow-up in the studies varied. Two studies (Cain et al., 2020; Wright et al., 2017a,b) utilized the four-week follow-up duration, which was the most popular. The remaining trials utilized a six-week (Cruz-Diaz et al., 2015; Hall et al., 2018a) and an eight-week (Kim et al., 2022a) follow-up protocol. All

studies incorporated proprioceptive input and resistance band training in one arm of their randomized controlled trials. However, there was some variation in the specific type of activity. The type and detailed description of each proprioceptive and resistance intervention is presented in Table 4.

3.2. Methodological quality

The maximum methodological quality score was seven points with scores ranging from five to seven (mean = 6.5, SD = 0.83). Only four of the included studies (80%) were considered of high methodological quality (≥ 6 points). The most common bias was the blinding of therapists and patients. Overall, the studies exhibited good quality. Table 5 lists the final score of the PEDro scale for each clinical study included in our systematic review/meta-analysis.

3.3. Risk of systematic error

Risk of bias assessment details of the included studies are shown in Fig. 2. In general, the risk of bias in our meta-analysis was low, apart from blinding of patients and therapists.

3.4. Proprioceptive training vs strength training on balance

Resistance training and proprioceptive training yielded similar results. Proprioceptive training was slightly better but not statistically significant (Fig. 3): (1) SEBT Ant: WMD: -0.71 , 95% CI = -3.64 to -2.21 , $I^2 = 89\%$, $P = .63$; (2) SEBT PM: WMD: -0.39 , 95% CI = -3.75 to -2.97 , $I^2 = 84\%$, $P = .82$; (3) SEBT PL: WMD: -1.76 , 95% CI = -4.83 to 1.30 , $I^2 = 74\%$, $P = .26$.

3.5. Proprioceptive training vs strength training on functional ability

Resistance training and proprioceptive training had similar effects on

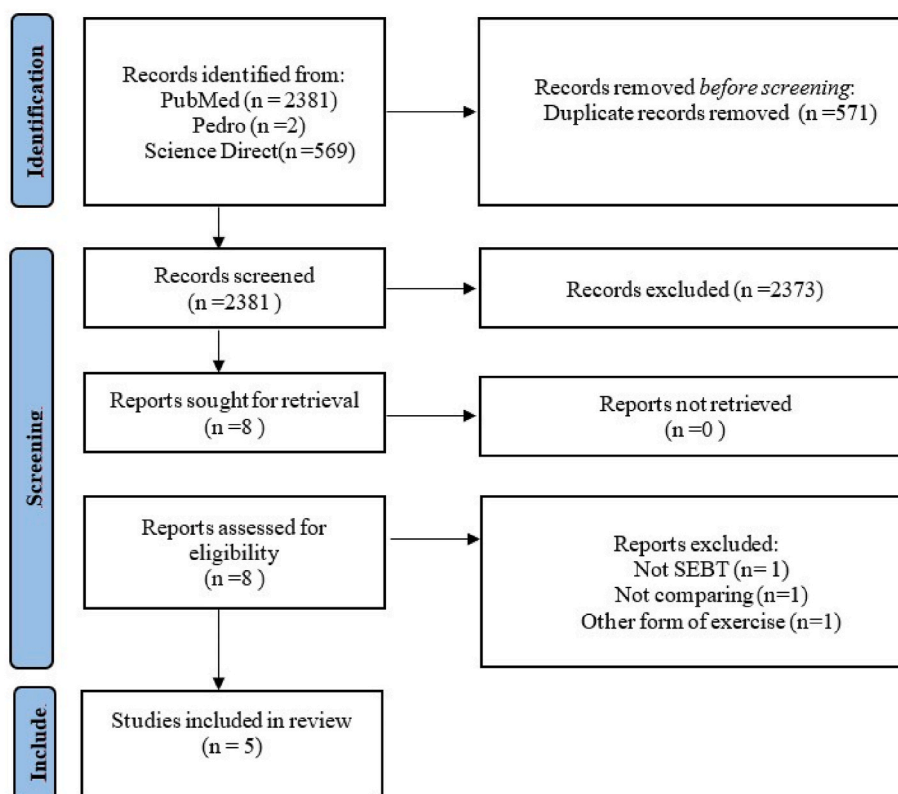


Fig. 1. PRISMA flow diagram depicting the results of the search strategy.

Table 3

Characteristics of participants of the included studies.

Study	Groups	Participants	Age	Gender, Men (Women)	Height (m)	Weight (kg)	BMI	Ankle sprains	Level of sporting participation
Cruz-Diaz et al. (2015)	G1	35	31.89 ± 10.52	15 (20)	1.71 ± 0.08	66.42 ± 11.75	22.65 ± 2.68	Unilateral	Athletes
	G2	35	28.83 ± 7.91	20 (15)	1.71 ± 0.09	69.34 ± 10.73	23.76 ± 2.80		
Cain et al. (2020)	G1	10	16.40 ± 0.97	8 (2)	1.79 ± 0.09	77.57 ± 21.54	–	Unilateral	Minimum of 5 h of moderate activity per week
	G2	12	16.42 ± 1.00	5 (7)	1.71 ± 0.10	65.75 ± 11.16	–		
	G3	10	16.20 ± 1.14	3 (7)	1.71 ± 0.02	67.99 ± 16.49	–		
	G4	11	16.45 ± 1.04	4 (7)	1.67 ± 0.14	67.17 ± 22.94	–		
	G1	13	23.5 ± 6.5	7 (6)	175.0 ± 0.08	72.8 ± 10.9	–		
Hall et al. (2018a)	G2	13	24.6 ± 7.7	8 (5)	173.2 ± 0.09	76.0 ± 16.2	–	Unilateral	NM
	G3	13	24.8 ± 9.0	6 (7)	175.5 ± 0.08	79.1 ± 16.	–		
	G1	20	22.60 ± 5.89	6 (14)	1.66 ± 0.15	70.25 ± 15.08	–		
Wright et al., 2017a,b	G2	20	21.45 ± 3.24	5 (15)	1.66 ± 0.87	76.38 ± 19.34	–	Unilateral	<1.5 h of moderate-vigorous physical activity per week
	G1	23	27.1 ± 5.8	14 (9)	173.7 ± 7.1	70.3 ± 8.8	–		
Kim et al., 2022a	G2	22	29.5 ± 10.7	11 (11)	171.8 ± 8.7	71.2 ± 9.4	–	Unilateral	Athletes
	G3	22	27.4 ± 8.3	16 (6)	170.5 ± 7.9	68.9 ± 9.2	–		

Table 4

Parameters of the treatments in the included clinical studies.

Study	Group	Intervention type	Intensity	Equipment	frequency	duration
Cruz-Diaz et al. (2015)	G1	Single and double limp stance	Moderate	Exercise mats, Dynair, Bosu, Minitramp, Foam roller	2–3 × /week	6 weeks
	G2	Single and multiple joint exercises covering different strength types, eccentric, concentric, and isometric, by closed and open chain exercises	Moderate	Resistance bands and Conventional fitness equipment	2–3 × /week	6 weeks
Cain et al. (2020)	G1	Single limb stance	5 trials changing direction every 10 s during each 40 s trial.	Biomechanical Ankle Platform System Board Intervention	3 × /week	4 weeks
	G2	Ankle plantar flexion, dorsiflexion, inversion, and eversion	3 sets 10 reps/set	Resistance band	3 × /week	4 weeks
	G3	Single limb stance Ankle plantar flexion, dorsiflexion, inversion, and eversion	5 trials changing direction every 10 s during each 40 s trial 3 sets 10 reps/set	Biomechanical Ankle Platform System Board Intervention Resistance band	3 × /week	4 weeks
Hall et al. (2018a)	G4	–	–	–	3 × /week	4 weeks
	G1	hop to stabilization, hop to stabilization and reach, hop-to-stabilization box drill, progressive single-limb-stance activities with eyes open, progressive single-limb-stance activities with eyes closed.	moderate	NM	3 × /week	6 weeks
	G2	Ankle plantar dorsiflexion, inversion, and eversion. heel raises slow-reversal PNF technique (concentric contraction of the antagonist muscle followed by a concentric contraction of the agonist muscle, against maximal manual resistance in a diagonal pattern)	3 sets 10 reps/set	Resistance band	3 × /week	6 weeks
Wright et al., 2017a,b	G3	Minute bicycle workout	20 min	Nm	3 × /week	6 weeks
	G1	Single limb stance	five 40 s sets of clockwise and counter- clockwise rotations (alternating direction every 10 s), 60 s of rest between sets.	Wobble Board	3 × /week	4 weeks
Kim et al., 2022a	G2	plantarflexion, dorsiflexion, inversion, and eversion	3 sets 10 reps/set	Resistance band	3 × /week	4 weeks
	G1	hop to stabilization, single limp stance	10 repetitions 30 s and 60 s, NM	Stable floor, mat, dynair, or busu	2 × /week	8 weeks
	G2	plantarflexion, dorsiflexion, inversion, and eversion	NM	Resistance band	2 × /week	8 weeks
	G3	–	–	–	2 × /week	8 weeks

Table 5

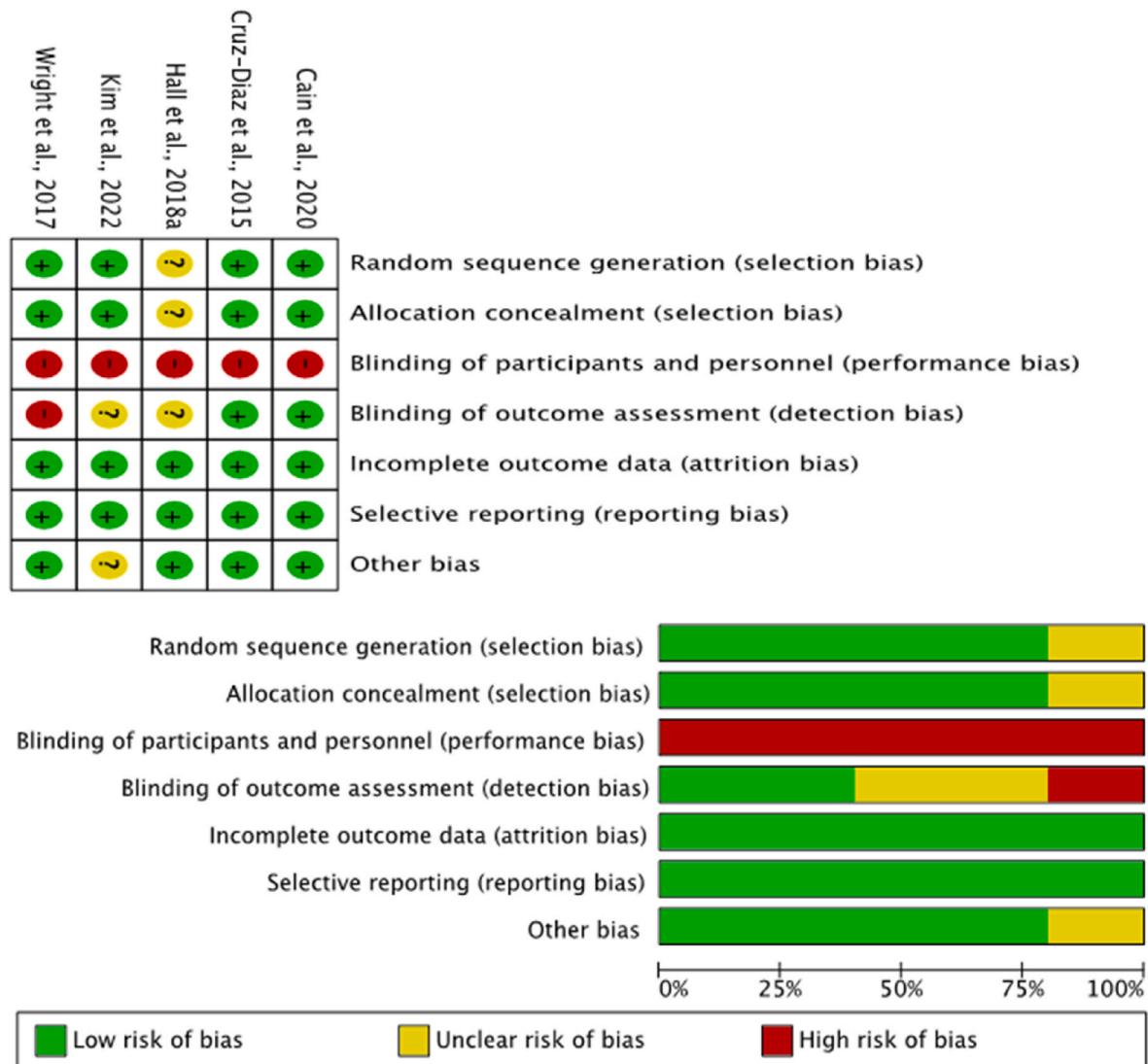
Methodological rating of randomized clinical trials using the PEDro baseline scale.

Study	1	2	3	4	5	6	7	8	9	10	Total
Cruz-Diaz et al. (2015)	Y	Y	Y	N	N	Y	Y	N	Y	Y	7/10
Cain et al. (2020)	Y	Y	Y	N	N	Y	Y	N	Y	Y	7/10
Hall et al. (2018a)	Y	N	Y	N	N	N	Y	N	Y	Y	5/10
Wright et al., 2017a,b	Y	Y	Y	N	N	N	Y	N	Y	Y	6/10
Kim et al., 2022a	Y	Y	Y	N	N	Y	Y	N	Y	Y	7/10

Y = yes, N = no.

1 = random allocation, 2 = hide allocation, 3 = basic comparability, 4 = subject blinding, 5 = therapist blinding, 6 = assessor blinding, 7 = results for 85%, 8 = intention-to-treat analysis, 9 = comparisons between groups, 10 = mean and variability data.

FAAM (Fig. 4): (1) FAAM Sport: WMD: 0.69, 95% CI = -3.68 to 5.05, I^2 = 22%, P = .76 (2) FAAM ADL: WMD: -2.26, 95% CI = -4.99 to 0.47, I^2 = 0%, P = .10. Proprioceptive training was statistically significantly better than resistance training in improving CAIT scores (Fig. 4): WMD: -2.21, 95% CI = -4.05 to -0.36, I^2 = 60%, P = .02, but these intervention results were not clinically significant (MDC, MCID score >3 point).

**Fig. 2.** Risk of systematic error charts.

3.6. Quality of evidence (GRADE)

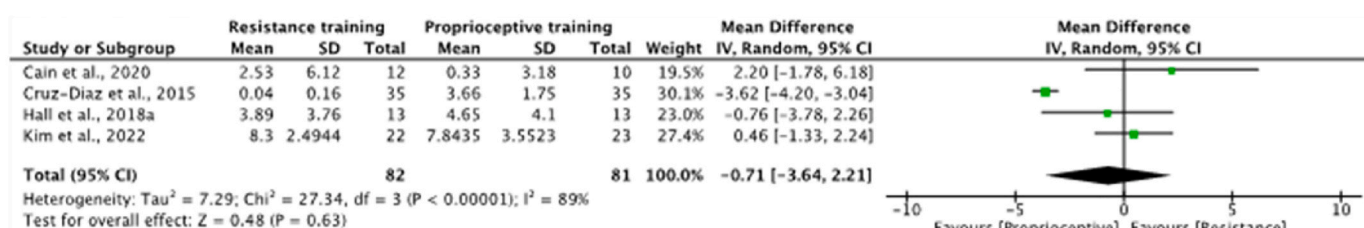
Table 6 lists the details of the GRADE assessment. Severe heterogeneity, as well as imprecision and publication bias reduced the quality level of the studies to a moderate to high degree regarding the overall effect of proprioceptive training and resistance training on balance and functional ability.

4. Discussion

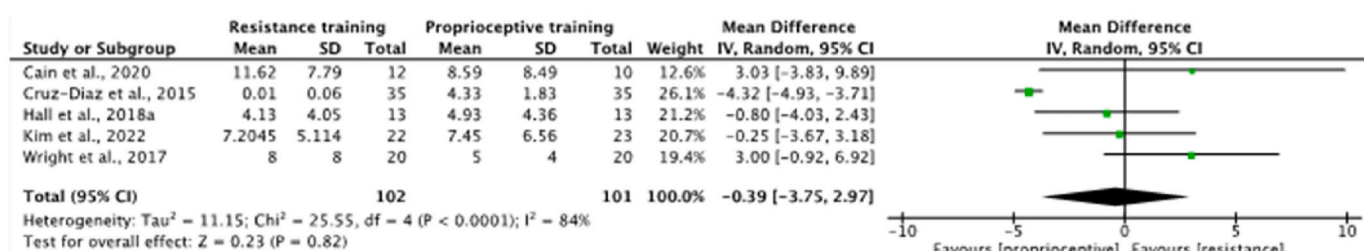
Our systematic review aimed to determine the relative efficacy of proprioceptive training and resistance training on CAI as measured by the SEBT, FAAM, and CAIT tests. Low to moderate evidence shows that proprioceptive training statistically differ from strength training in terms of functional ability, instability, or dynamic balance. However, we found that proprioceptive training statistically significantly improves CAIT scores more than resistance training. It is important to note that the WMDs were lower than their MDC values, thus the observed statistically significant improvement in CAIT did not reflect a discernible change.

Regarding the above, the findings show the potential of both interventions as neither was superior to the other in clinical practice or in a sport setting. Moreover, the included studies found beneficial results for outcomes related to functional ability, instability, and balance. We

SEBT- ANT



SEBT- PM



SEBT- PL

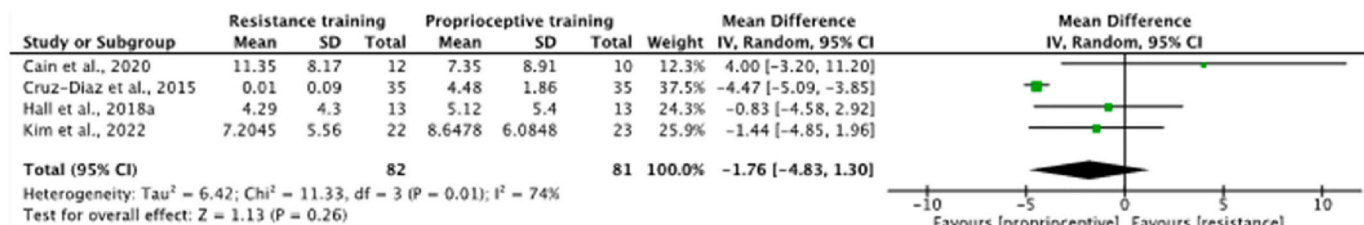


Fig. 3. Comparison of proprioceptive training and strength training on balance.

concluded, therefore, that clinicians can use either method with similar rehabilitation outcomes in patients with CAI.

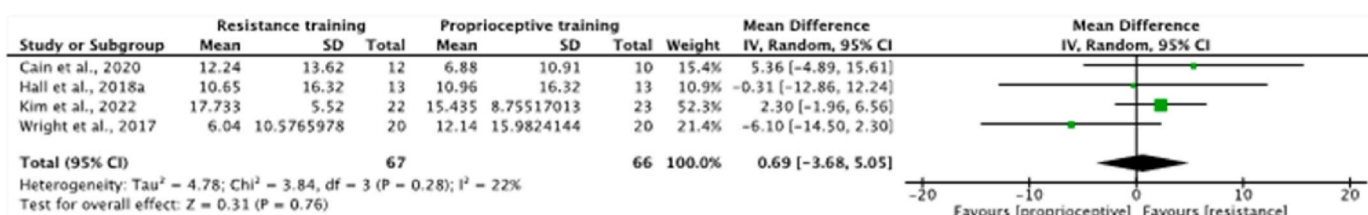
To our knowledge, our work is the first meta-analysis to analyze and compare the effect of proprioceptive training with resistance training on functional ability, instability, and dynamic balance in CAI. After a thorough search, no systematic reviews were found that studied the effect of proprioceptive training and strength training. However, we found studies that compared balance exercises and strength training individually on functional ability, instability, and dynamic balance. Such studies included that of Luan et al. (2021), who compared strength training for CAI on balance and patient-reported outcomes, Postle et al. (2012), who investigated the efficacy of proprioceptive exercises for ankle ligament injury in adults, Ahern et al. (2021a), who studied the effects of functional rehabilitation on Performance of the SEBT among recreational athletes with CAI, and Mollà-Casanova et al. (2021), who focused on the effects of balance training on functional ability, ankle instability, and dynamic balance. Based on the results of these evaluations, we were unable to directly compare the effectiveness of the two

strategies of interest here.

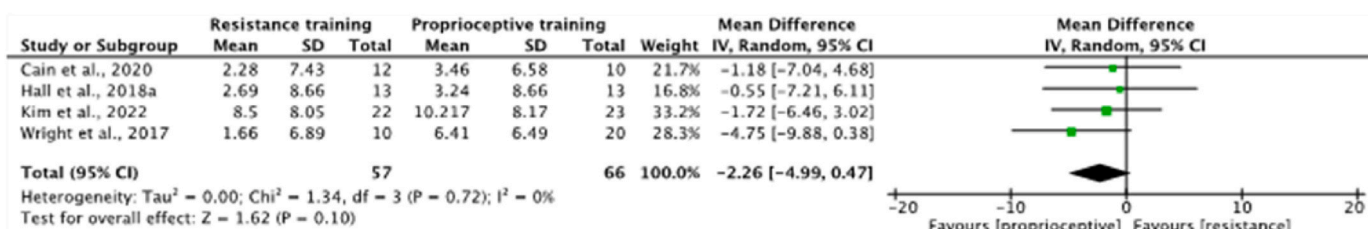
The main deficits associated with CAI affect proprioception, neuromuscular control, strength, and postural control (Holmes & Delahunt, 2009). Previous studies have shown that patients with CAI demonstrated a significant lower extremity strength weakness compared to individuals without CAI (Cho et al., 2019; Mattacola & Dwyer, 2002), which implies poor postural control in individuals with CAI. Additionally, existing research has also reported that patients with CAI rely more on their hip and not their ankle during postural control tasks, due to the inability of the muscles surrounding the ankle to produce the required strength (Doherty et al., 2016). A recent systematic review showed that individuals with CAI have ankle inversion and eversion strength deficits (Khalaj et al., 2020). Consequently, in our meta-analysis we tried to compare ankle muscle strengthening exercises with resistance bands to proprioceptive exercises.

Proprioception mainly concerns the positional sense derived from the mechanoreceptors (Macefield, 2021) and is a general term for a particular component of tactile perception of joint movement including

FAAM-Sport



FAAM-ADL



CAIT

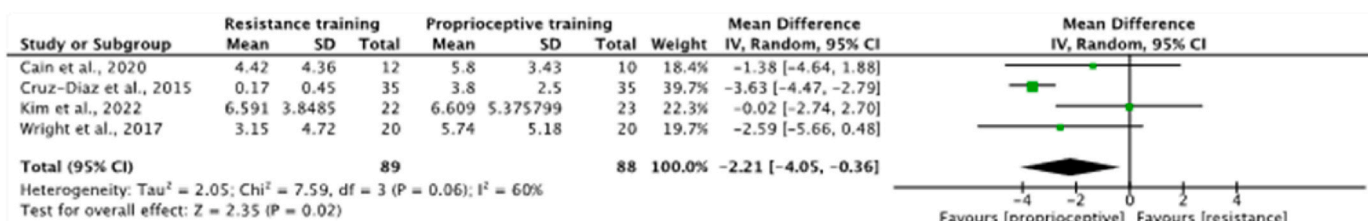


Fig. 4. Comparison of proprioceptive training and strength training on function.

awareness of kinesthesia and joint position sense (Lephart et al., 1997; Macefield, 2021). For this reason, proprioceptive deficits and loss of joint position are associated with loss of balance (SEBT) and functional ability (CAIT, FAAM).

Our meta-analysis showed that, when compared with resistance band training, proprioceptive exercises were superior in terms of balance as measured by SEBT. However, due to high variation of proprioceptive exercise programs and different elastic tube exercises protocols all three SEBT scores were highly heterogeneous ($I^2 > 50\%$). Moreover, results favoring proprioceptive exercises were not statistically significant, which is the reason why they were not mentioned before.

Additionally, CAIT scores revealed a similar outcome in favor of proprioceptive exercises with lower heterogeneity than SEBT scores. Furthermore, CAIT scores were statistically significant but not clinically significant. Regarding FAAM scores, proprioceptive exercises were superior in ADL, and resistance exercises were superior in sports activities. Both scores were not statistically significant, but they exhibited low heterogeneity. The results of studies using the FAAM-Sport and FAAM-ADL show different patterns of outcomes. One study reported better

results in FAAM-Sport for resistance training and better results in FAAM-ADL for proprioceptive training (Cain et al., 2020). Consequently, our results vary.

Our review has identified several major methodological limitations to the current evidence. Firstly, the number of randomized controlled trials examining our comparison was relatively small ($n = 5$). Secondly, the included sample was relatively small, which substantially impacted the difference between pre- and post-intervention scores. Given the relatively small number of participants, studies could have better blinded them and avoided performance bias. Another important limitation is the large heterogeneity between studies, which, however, may be explained because different clinical trials used different proprioceptive and resistance exercise protocols (different length of elastic tubes, equipment in proprioceptive protocols and length of trial) in patients with varying characteristics. Only two studies reported the length and type of elastic band used (Cain et al., 2020; Wright et al., 2017a,b). The resistance was determined using the methods of Kaminski et al. (2003) by calculating 70% of the resting length of the band, then adding this distance to the resting length of the band. The equipment used in the

Table 6

GRADE evidence for Proprioceptive training versus resistance training in balance.

Number of studies	Risk of Bias ^a	Inconsistency ^b	Indirectness of Evidence ^c	Imprecision ^d	Publication Bias ^e	Quality of Evidence	MD or SMD (95% CI)
Proprioceptive training versus resistance training in balance (SEBT)							
SEBT-Ant (n = 163)	No	Very Serious ($i^2 = 89\%$)	No	No	Yes	Low	–0.71 [–3.64, 2.21]
SEBT-PM (n = 203)	No	Very Serious ($i^2 = 84\%$)	No	No	Yes	Low	–0.63 [–2.17, 0.91]
SEBT-PL (n = 163)	No	Serious ($i^2 = 74\%$)	No	No	No	Moderate	–0.39 [–3.75, 2.97]
FAAM-ADL (n = 123)	No	No ($i^2 = 0\%$)	No	Serious	No	Moderate	–0.54 [–1.87, 0.79]
FAAM-Sport (n = 129)	No	No ($i^2 = 22\%$)	No	Serious	No	Moderate	–1.76 [–4.83, 1.30]
CAIT (n = 177)	No	Serious ($i^2 = 60\%$)	No	No	No	Moderate	–0.83 [–2.46, 0.80]
							–2.26 [–4.99, 0.47]
							–0.27 [–0.63, 0.09]
							0.69 [–3.68, 5.05]
							0.04 [–0.35, 0.43]
							–2.21 [–4.05, –0.36]
							–0.72 [–1.67, 0.22]

GRADE = Grading of Recommendations Assessment, Development, and Evaluation; MD = mean difference; SMD = standardized mean difference.

^a “No” = most information is from results at low risk of bias; “Serious” = crucial limitation for one criterion or some limitations for multiple criteria sufficient to lower confidence in the estimate of effect; “Very serious” = crucial limitation for one or more criteria sufficient to substantially lower confidence in the estimate of effect.

^b “Serious” = $I^2 > 40\%$; “Very serious” = $I^2 > 80\%$.

^c No indirectness of evidence was found in any study.

^d Based on sample size. “Serious” = $n < 142$ subjects; “Very serious” = $n < 142$ and the estimated effect is little or absent.

^e Based on funnel plots. No publication bias was found. Funnel plots are not shown because the number of trials was less than 10.

proprioception groups also varied considerably. Only two studies used some common equipment such as mats, DYNAIR, and BOSU (Cruz-Diaz et al., 2015; Kim et al., 2022a). Cruz-Diaz et al. (2015) reported using mini tramp and foam roller. Cain et al. (2020) reported using Biomechanical Ankle Platform System (BAPS) Board, Wright et al. (2017a,b) used a wobble board, and Hall et al. (2018a) did not mention any equipment. Follow-up time was significantly different among studies. Cruz-Diaz et al. (2015) and Hall et al. (2018a) reported a six-week intervention protocol, Cain et al. (2020) and Wright et al. (2017a,b) a four-week protocol, and Kim et al. (2022a) an eight-week intervention protocol. The literature was largely based on younger aged groups, with the mean age of the review being 22. As a result, it is impossible to say if older adults could benefit from treatment. It is important to mention that older adults have worse joint position sensing than younger people and therefore might benefit more from proprioceptive training.

Considering the above as well as our results, more high-quality randomized controlled trials are needed to evaluate the efficacy of resistance training with elastic bands compared to proprioceptive training in self-reported outcomes in patients with CAI. The five studies compared in our review may not be adequate to establish a firm conclusion. It is clinically important to consider the age of the patient when planning future trials. Future studies should investigate whether the combination of both interventions is more beneficial than either alone. It is also advised to conduct more research comparing proprioceptive exercise and resistance exercise in general. Additional trials are yet needed to evaluate which intervention is better suited for those who experience repeated injuries during sports activities or daily life tasks. Future research should also include an adequate follow-up period to establish long-term effects as the chronic state of an ankle instability is constantly changing. In conclusion, it is important to find the appropriate dosage and duration for interventions to create treatment guidelines and allow clinicians to clearly instruct their patients.

The objective of our review, despite multifaceted nature of CAI, is to better understand how it develops into chronic damage and to prevent further harm.

5. Conclusion

According to the findings of our meta-analysis, proprioceptive training achieved similar beneficial effects to those of resistance training in balance and patient-reported outcome measures. Nevertheless, our

results did not exceed the minimal detectable change in people with CAI. We noted that proprioceptive training slightly better improved all the above-mentioned results but was not statistically significant, suggesting that clinicians should prioritize proprioceptive interventions over resistance training with elastic bands for improving balance and symptoms in CAI. The small number and lack of long-term follow-up studies prevent us from drawing firm conclusions. More high-quality studies comparing the two interventions are needed to provide evidence on which of the two is the gold standard in the management of CAI.

Ethical approval

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Declaration of competing interest

None.

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